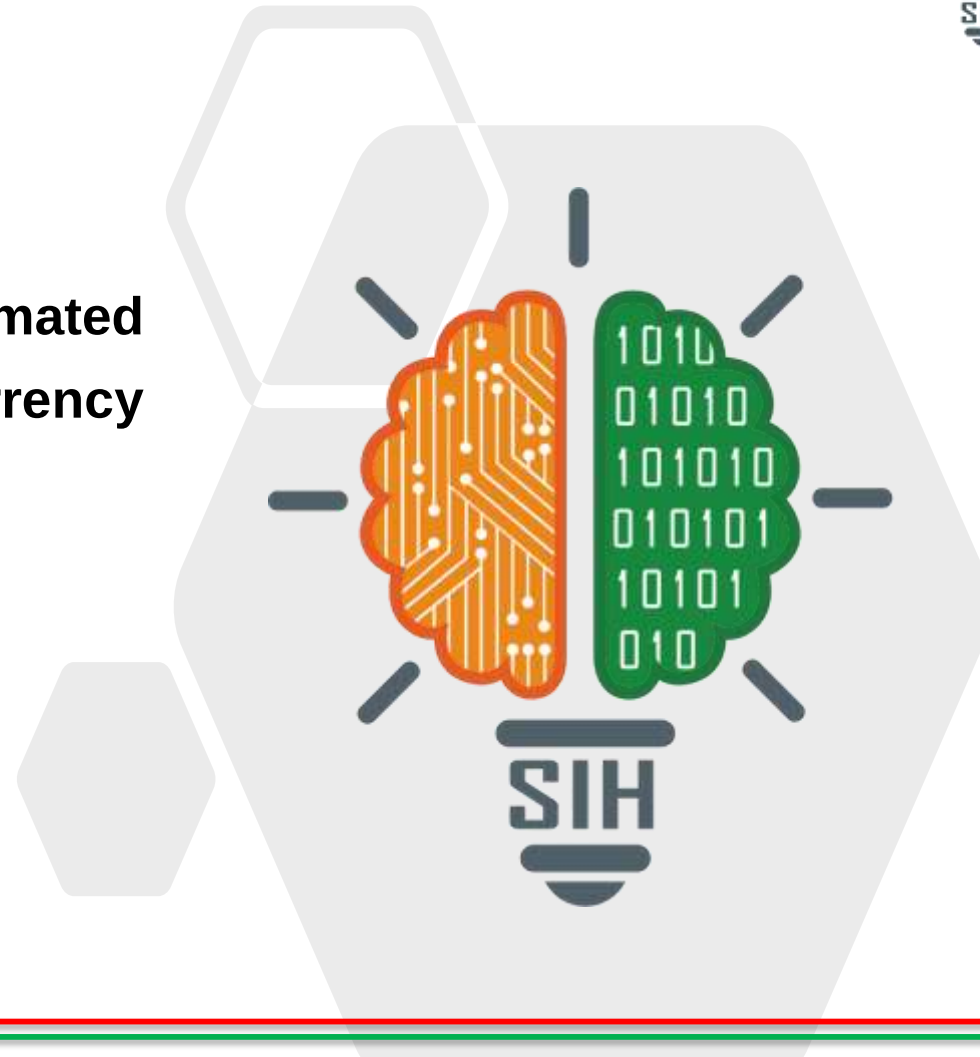


SMART INDIA HACKATHON 2026



- Problem Statement ID – SIH26182
- Problem Statement Title- Automated Attribution of Unknown Cryptocurrency Wallets to Nearest VASPs
- Theme- Blockchain & Cybersecurity
- PS Category- Software
- Team ID- 177141
- Team Name- Genesis-Blockers



THEMES & TECHNOLOGIES

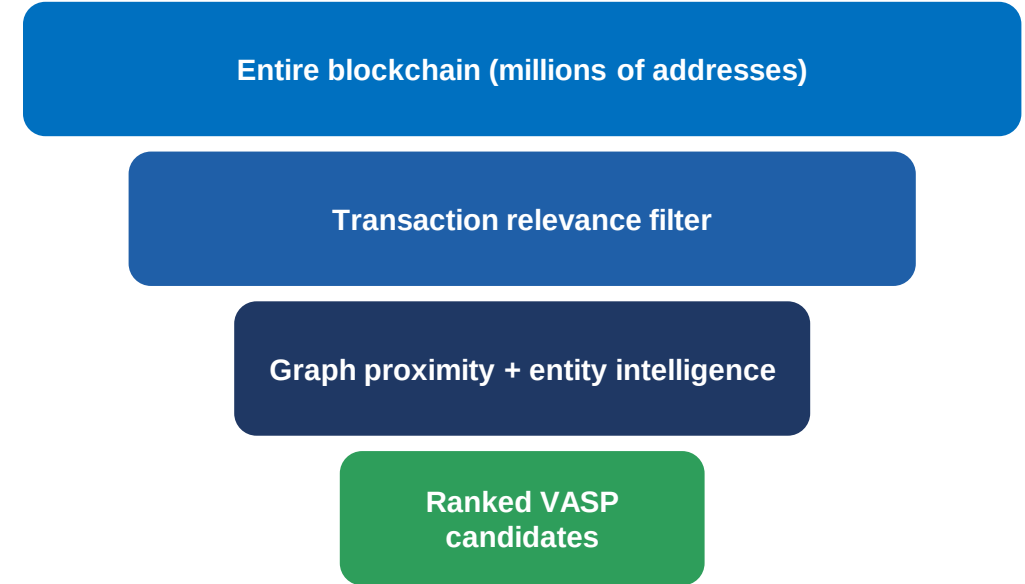
Blockchain Intelligence

Transaction Graph
Analysis

Evidence-Based
Attribution

❖ Proposed Solution

- **Problem:** Pseudonymous addresses and multi-hop transactions hide real VASP ownership.
- **Core Idea:** Progressive Filtering - narrow the search using transaction, graph and entity evidence.
- **Outcome:** A huge search space becomes a few evidence-ranked VASP candidates.
- **Why it works:** Label lookup misses links several hops away; a graph exposes them.



INNOVATION & USP

Search-space reduction

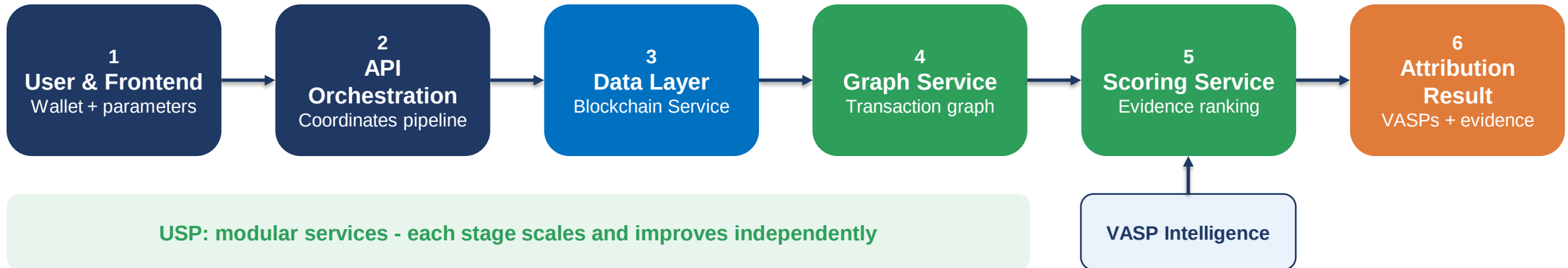
Filters first, scores later - not brute-force label matching.

Multi-hop graph evidence

Finds VASP links several transactions away.

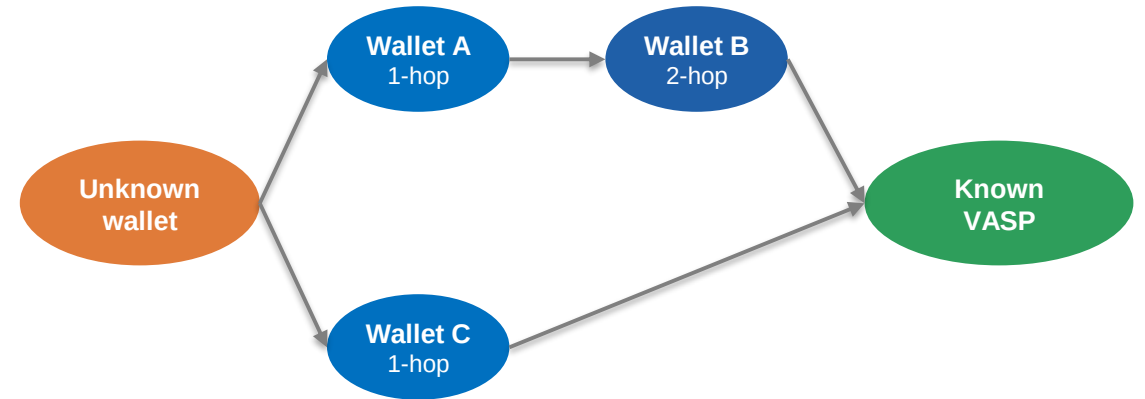
Explainable by design

Every result shows Who, Why and How.

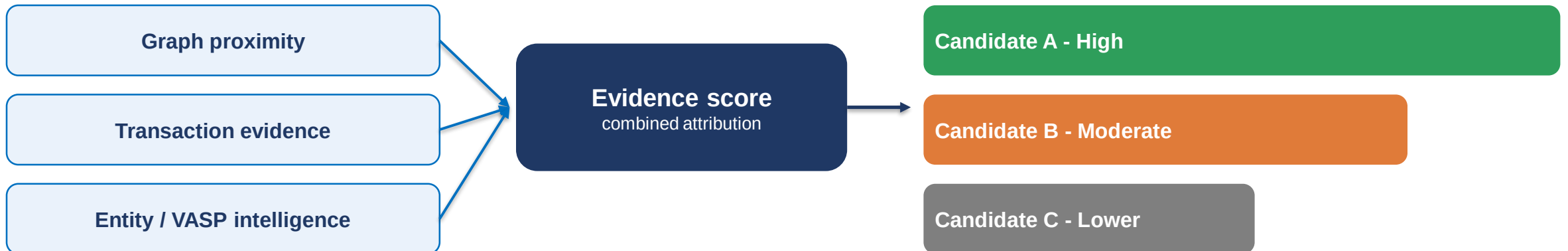


- **API Orchestration:** Coordinates the pipeline and service-to-service communication.
- **Graph Service:** Turns transfers into a traversable graph - wallets are nodes, transactions are edges (value, time, tx ID).
- **Scoring Service:** Ranks candidate VASPs by accumulated graph, transaction and entity evidence.
- **Data & VASP Intelligence:** Normalized blockchain data plus known-entity enrichment.

- **Neighborhood Discovery:** Hop-based traversal from the unknown wallet - 1-hop counterparties, 2+ hop peers.
- **Why it works:** Attribution signals often sit several transactions away.
- **Risk:** Large, noisy graphs. Fix: relevance filtering before scoring.



EVIDENCE AGGREGATION MODEL



IMPACT AND BENEFITS

WHO?

Candidate VASP

WHY?

Supporting transaction & graph evidence

HOW?

Multi-hop path linking wallet to VASP

System output: ranked attribution

Candidate VASP A - Strong evidence

Candidate VASP B - Moderate evidence

Backed by: tx hashes, graph paths, known-entity info

- **Automated:** cuts manual effort tracing multi-hop transaction networks.
- **Graph-driven:** uses relationships, not just direct address labels.
- **Explainable:** every attribution is backed by paths and evidence.
- **Auditable:** results can be reviewed and verified end to end.

Transparent, auditable attribution - answering WHO, WHY and HOW

- **GitHub:** <https://github.com/Genesis-Blockers/Genesis-Blockers>
- **Research area:** Blockchain multi-hop transaction graph analysis.
- **Approach:** Evidence-based VASP attribution using graph proximity and transaction evidence.

